

Unsupervised Learning Multi-Class Classification using Fashion MNIST Data

1. Introduction

This report uses statistical learning analysis of the Fashion MNIST dataset, integrating unsupervised learning and multi-class classification models. After applying PCA (principal component analysis) to reduce dimensionality of the 784 pixel features, clustering algorithms like k-means and hierarchical clustering were evaluated by looking at the silhouette scores as well as the dominant-label analysis. Results consistently indicate weak natural clustering structure among Fashion-MNIST categories. Because these clustering algorithms are not the most successful when finding meaningful distinctions between different groups, we are using multi-class classification models.

The 4 supervised multi-class classification models we will be trained to predict clothing categories are: multinomial logistic regression, k-nearest neighbors (KNN), support vector machine (SVM), and random forest. Among them, the nonlinear SVM classifier achieved the lowest test error, highlighting the importance of nonlinear decision boundaries for visual pattern recognition tasks.

The Fashion-MNIST dataset contains 70,000 grayscale images, 60,000 training and 10,000 testing, each representing one of ten clothing categories, such as shirts, T-shirts, dresses, coats, sandals, sneakers, etc. Each 28×28 image was flattened into a 784-dimensional vector, yielding a high-dimensional classification problem with substantial overlap between visually similar classes. For example, Shirt vs T-shirt vs Pullover as they often appear similar in pictures which makes it difficult to differentiate.

This project has two main objectives:

1. **Unsupervised Learning:**
 - Exploring if fashion dataset has clusters emerge.
 - Use silhouette scores to assess if the data has separable clusters.
 - Identify dominant labels and evaluate what potential clusters could represent.
2. **Multi-class Classification Models:**
 - Train 4 multi-class classification models: logistic regression, KNN, SVM, and random forest.
 - Compare their accuracy on the testing dataset.
 - Analyze which models best capture nonlinear structure.

2. Data Pre-Processing

Before conducting any statistical analysis, several preprocessing steps were necessary to prepare the data.

2.1 Flattening and Feature Extraction

Each 28×28 image was unfolded into a 784-dimensional vector and scaled to ensure uniformity. Because modeling the Fashion-MNIST data is difficult due to high-dimensional data, Principal Component Analysis (PCA) was used for dimensionality reduction. Many machine learning algorithms, especially clustering and distance-based models like KNN perform poorly in such high dimensions due to the curse of dimensionality, in which dimensionality reduction is necessary.

2.2 Principle Component Analysis (PCA)

PCA significantly reduces computational cost while preserving essential structure. PCA was performed on the training data:

- Images were centered and scaled.
- Eigenvalues were examined to determine the cumulative variance explained.
- The first 84 principal components were retained, capturing ~90% of total variance.
- These 84 components served as input for both clustering and classification.

2.3 Frequency Tables for Outcome Variable

- Training Data (60,000 total)
- Each label appears 6,000 times, demonstrating perfect class balance.
- Testing Data (10,000 total)
- Each label appears 1,000 times.

Frequency tables for the training and testing labels confirmed that both datasets were perfectly balanced, with equal representation from all 10 classes.

```
## y_train
## 0 1 2 3 4 5 6 7 8 9
## 6000 6000 6000 6000 6000 6000 6000 6000 6000 6000

prop.table(train_freq)

## y_train
## 0 1 2 3 4 5 6 7 8 9
## 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1

test_freq

## y_test
## 0 1 2 3 4 5 6 7 8 9
## 1000 1000 1000 1000 1000 1000 1000 1000 1000 1000

prop.table(test_freq)

## y_test
## 0 1 2 3 4 5 6 7 8 9
## 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1 0.1
```

3. Unsupervised Learning

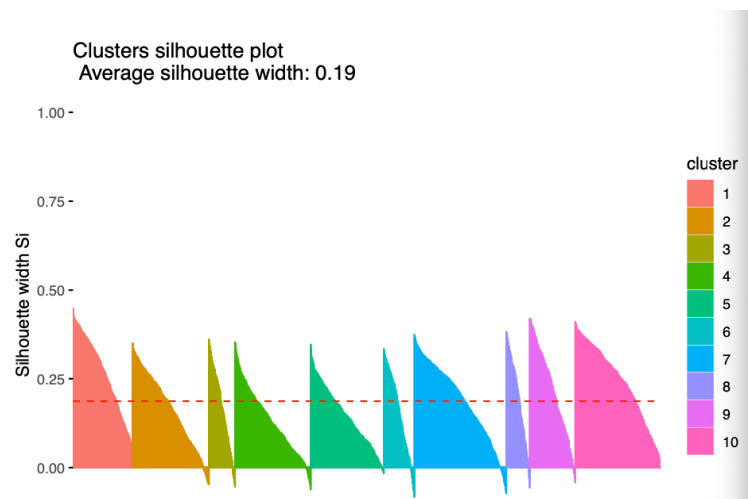
The first analytical task involved exploring the unstructured patterns in the dataset using clustering algorithms. A full pairwise distance matrix for 60,000 images is computationally infeasible, so clustering was performed on a representative subsample of 3,000 PCA-transformed images. Clustering methods **k-means** and **hierarchical clustering** were evaluated using silhouette scores and dominant-label inspection. Across all methods, clusters displayed weak structure and significant overlap between clothing categories.

Each clustering method was evaluated according to:

1. **Silhouette statistic** → determines clustering validity.
2. **Dominant label per cluster** → identifies interpretability.
3. **Overlap among labels** → evaluates separation quality.

Silhouette Statistic to find optimal number of clusters

##	cluster	size	ave.sil.width
## 1	1	507	0.26
## 2	2	649	0.16
## 3	3	223	0.17
## 4	4	644	0.13
## 5	5	622	0.14
## 6	6	261	0.15
## 7	7	781	0.19
## 8	8	201	0.20
## 9	9	387	0.22
## 10	10	725	0.24

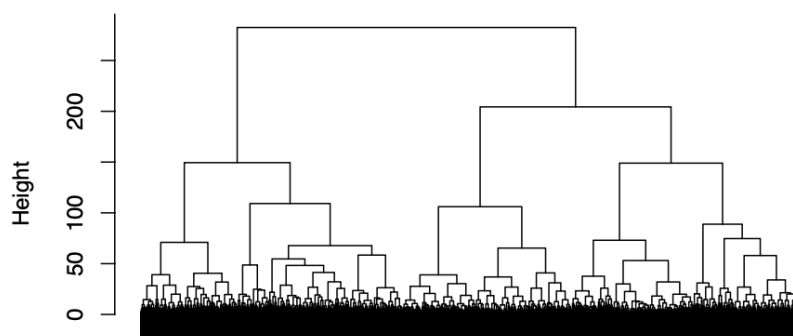


```
##      y_sub
##      0  1  2  3  4  5  6  7  8  9
## 1    0  0  0  0  0 113  1 358 24 11
## 2 302 26  9 149 81  0 80  0  2  0
## 3    1  0  2  1  3  0  4  0 212  0
## 4   55 18 38  50 22 292 78 42 38 11
## 5 119  6 158 51 95  3 169  0 18  3
## 6    0  0  0  0  0 23  0  2  1 235
## 7   17  6 299  3 280  0 151  0 24  1
## 8    4  0  1  0  2  0  8  1 185  0
## 9    0  0  0  0  0 51  0 77  7 252
## 10  20 432  1 248 15  0  3  0  6  0
```

```
##      1  2  3  4  5  6  7  8  9 10
## "7" "0" "8" "5" "6" "9" "2" "8" "9" "1"
```

Hierarchical Clustering

Hierarchical Clustering Dendrogram



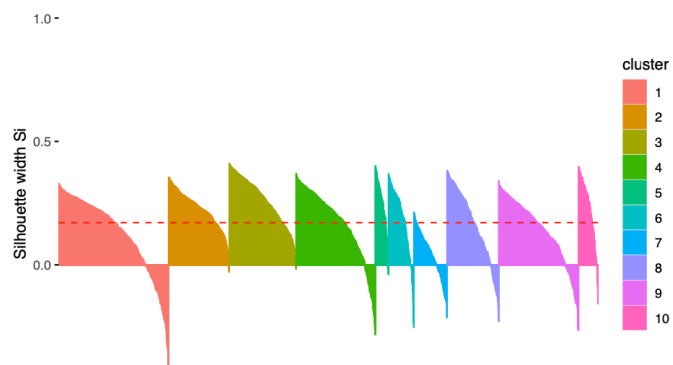
```
##      y_sub
## hc_clusters 0  1  2  3  4  5  6  7  8  9
## 1    18  5 220  2 212  0 121  0 33  0
## 2   83  4  73 30 31  0 107  0  9  0
## 3    0  0  0  0  0 129  0 225 15  3
## 4    9 254  0 148  3  0  3  0 23  0
## 5    0  0  0  0  0  5  0  0  0 68
## 6    1  0  1  0  1  0  2  0 136  0

##      7  1  0  0  0  1 145  2 19  7  9
##      8  0  0  0  0  0 11  0 64  0 213
##      9 199 19  5 117 45  0 57  0  1  0
##     10  4  0  0  0  1  0  9  0 97  0
```

```
##      1  2  3  4  5  6  7  8  9 10
## "2" "6" "7" "1" "9" "8" "5" "9" "0" "8"
```

Clusters silhouette plot
Average silhouette width: 0.17

```
##      cluster size ave.sil.width
## 1          1  611          0.13
## 2          2  337          0.23
## 3          3  372          0.26
## 4          4  440          0.17
## 5          5   73          0.25
## 6          6  141          0.18
## 7          7  184          0.04
## 8          8  288          0.16
## 9          9  443          0.14
## 10         10  111          0.22
```

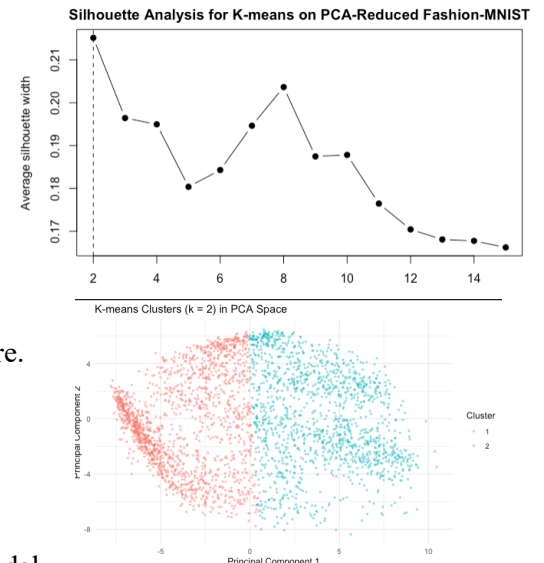


3.1 K-Means Clustering

K-means clustering was applied with $k = 10$ to match the number of known clothing categories. The resulting clusters show identifiable dominant labels but also substantial mixing. The average silhouette width was approximately 0.17, suggesting weak separation. For example, trouser images tended to cluster together more strongly than shirts or pullovers, which exhibited considerable overlap due to visual similarity.

Silhouette Statistic

- Looking at the silhouette score plot, the optimal number of clusters looks like $k=2$, with an average silhouette width of ~ 0.19 . This is extremely low, indicating very weak natural cluster structure in the Fashion MNIST data. However, because the dataset contains ten known clothing categories, we examined k-means with $k=10$ to evaluate cluster interpretability and identify dominant labels. The $k=10$ solutions show substantial label mixing, confirming that unsupervised learning does not recover the true classes.
- Number of clusters: **$k = 10$** , chosen to correspond with the ten true clothing categories.
- Average silhouette width: ~ 0.17** , indicating very weak cluster structure.
- Plotting the data essentially shows 2 “blob” clusters, and in terms of analysis doesn’t mean anything significant.



Dominant Labels in Each Cluster

From the cluster $\times$ label table, each cluster had a dominant class (Pullover, Sandal, Ankle boot), but many clusters included substantial label mixing.

Interpretation

- Trousers (label 1) formed more compact clusters.
- Shirt-like categories (0, 2, 4, 6) were highly intermixed.
- K-means failed to uncover strong natural grouping.

3.2 Hierarchical Clustering

Hierarchical clustering with Ward’s linkage provided a dendrogram revealing some high-level separation but highly entangled lower-level structure. The silhouette width (~ 0.17) was nearly identical to k-means, reinforcing the conclusion that natural clusters in Fashion-MNIST are weakly defined. Many clusters included mixed labels, particularly among shirt-like categories.

Hierarchical clustering using Ward’s D2 linkage produced:

- Silhouette width: ~ 0.17** (almost identical to k-means)
- A dendrogram showing some coarse separation but heavy mixing at lower branches.

Interpretation

The hierarchical tree indicates that:

- High-level splits separate items like shoes vs tops.

- Lower-level splits fail to distinguish similar tops (T-shirt, Shirt, Pullover).

Like K-means, hierarchical clustering confirms that the dataset is **not naturally cluster able**.

3.5 Summary of Unsupervised Findings

Overall, all clustering methods indicated that Fashion-MNIST does not possess well-separated natural clusters in PCA space. This aligns with human intuition: many clothing categories are defined by subtle stylistic differences rather than sharp structural boundaries. As a result, effective classification requires supervised learning methods capable of modeling nonlinear decision boundaries.

Do the clusters help separate labels? **No**.

Clustering algorithms demonstrated:

- Weak separation between classes
- Heavy overlap among shirts, T-shirts, pullovers, and coats
- Silhouette scores < 0.20
- Inconsistent cluster purity

Fashion-MNIST **does not exhibit strong natural clusters** in PCA-reduced space.

Therefore, **multi-class classification is required** to achieve good predictive performance.

4. Multi-Class Classification Models

Implementing multi-class classification models using the PCA-transformed data, the models evaluated were:

- Multinomial Logistic Regression
- k-Nearest Neighbors ($k=5$)
- Support Vector Machine with an RBF kernel
- Random Forest.

All models were trained on the 60,000-image training set and evaluated on the 10,000-image test set. Performance was assessed using misclassification error rates and examination of confusion matrices.

4.1 Multinomial Logistic Regression

Multinomial logistic regression served as a linear baseline model. It achieved a test error rate of 0.1583. Most misclassifications occurred among shirt-like categories, where linear boundaries fail to capture subtle visual distinctions. Nonetheless, its performance provides a useful benchmark for more flexible models.

##	True										
## Predicted	0	1	2	3	4	5	6	7	8	9	
##	0	815	1	12	38	1	0	167	0	4	0
##	1	6	968	1	26	1	0	4	0	0	1
##	2	16	9	762	17	87	0	111	0	10	0
##	3	47	15	10	866	33	0	39	0	4	0
##	4	9	1	121	24	791	0	128	0	5	0
##	5	2	2	0	1	1	901	1	42	9	15
##	6	88	4	84	25	83	1	535	0	27	0
##	7	1	0	1	0	0	74	0	908	8	41
##	8	16	0	9	3	3	6	14	1	928	0
##	9	0	0	0	0	0	18	1	49	5	943

4.2 k-Nearest Neighbors (k = 5)

The k-Nearest Neighbors classifier (k=5) produced a test error of 0.1352, outperforming logistic regression. Because KNN makes decisions based on local neighborhoods, it can model some nonlinear structure present in image data. However, KNN struggles in high-dimensional spaces, even with PCA, and predictions can be computationally expensive.

##	True										
## Predicted	0	1	2	3	4	5	6	7	8	9	
##	0	847	4	15	27	3	0	182	0	1	0
##	1	1	968	0	7	0	0	1	0	0	0
##	2	14	3	763	12	80	1	89	0	10	0
##	3	15	17	13	901	22	1	18	0	2	0
##	4	6	2	129	33	806	0	77	0	5	0
##	5	0	0	0	0	0	867	0	10	1	4
##	6	107	6	77	19	86	2	623	0	3	0
##	7	2	0	0	0	0	78	0	938	4	32
##	8	8	0	3	1	3	4	10	0	971	0
##	9	0	0	0	0	0	47	0	52	3	964

4.3 Support Vector Machine (RBF Kernel)

The Support Vector Machine (SVM) with an RBF kernel achieved the lowest test error at 0.0938. This model successfully captured complex nonlinear boundaries between classes. SVM is well-suited to PCA-transformed data, as the reduced dimensionality mitigates computational demands while preserving discriminative structure. Confusion matrices showed persistent misclassification among shirts, T-shirts, and pullovers—categories that are challenging even for deep learning models.

##	True										
## Predicted	0	1	2	3	4	5	6	7	8	9	
##	0	860	4	12	22	0	0	108	0	6	0
##	1	0	981	1	7	0	0	0	0	0	0
##	2	18	2	829	10	71	1	71	0	4	0
##	3	16	10	11	921	18	0	21	0	0	0
##	4	1	0	79	18	847	0	45	0	2	0
##	5	0	1	1	0	1	966	0	11	2	4
##	6	101	2	65	22	61	0	750	0	6	0
##	7	0	0	0	0	0	21	0	962	1	29
##	8	4	0	2	0	2	2	4	0	979	0
##	9	0	0	0	0	0	10	1	27	0	967

4.4 Random Forest

The Random Forest classifier achieved a test error rate of 0.1302, comparable to KNN. While Random Forest models excel with tabular data, their performance on high-dimensional continuous features can be limited. Nevertheless, they offer robustness and interpretability, making them valuable components of the model comparison.

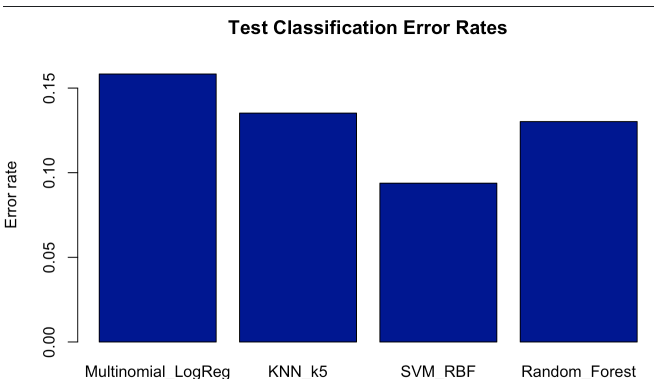
##	True									
## Predicted	0	1	2	3	4	5	6	7	8	9
##	0	851	4	11	29	1	0	169	0	2
##	1	0	960	0	6	1	0	1	0	0
##	2	16	9	803	12	66	1	95	0	9
##	3	33	22	10	907	27	1	25	0	3
##	4	4	1	105	24	847	0	76	0	4
##	5	2	0	1	0	0	915	0	38	12
##	6	83	4	61	20	53	0	620	0	7
##	7	1	0	0	0	0	51	0	899	4
##	8	10	0	9	2	5	7	14	0	957
##	9	0	0	0	0	0	25	0	63	2

4.5 Summary of Classification Results

Comparing all four supervised models, SVM demonstrated the best performance, followed by Random Forest and KNN, with logistic regression performing the worst. These results indicate that nonlinear models are essential for capturing the decision boundaries between visually similar classes in Fashion-MNIST.

## Multinomial_LogReg	KNN_k5	SVM_RBF	Random_Forest
## 0.1583	0.1352	0.0938	0.1302

Model	Test Error
SVM (RBF)	0.0938
Random Forest	0.1302
KNN (k = 5)	0.1352
Logistic Regression	0.1583



5. Conclusion

This project demonstrated the challenges of unsupervised learning on high-dimensional visual data and the strengths of supervised learning approaches for multi-class prediction. Clustering methods revealed weak intrinsic structure among the ten Fashion-MNIST categories, while supervised models—especially the SVM—achieved strong predictive accuracy.

This project highlights the contrast between unsupervised and supervised learning performance on complex image datasets. Unsupervised clustering methods—K-means and hierarchical clustering—all produced weakly separated clusters due to overlapping visual features among clothing categories. Silhouette statistics consistently confirmed that Fashion-MNIST possesses no strong natural clustering structure.

In contrast, supervised learning models—particularly the nonlinear SVM—achieved strong predictive accuracy. The SVM achieved the lowest test error (9.38%), outperforming logistic regression, KNN, and random forest. These results illustrate the necessity of nonlinear modeling techniques when dealing with subtle visual distinctions. Overall, this analysis demonstrates that while clustering provides limited insight into Fashion-MNIST structure, supervised learning—especially nonlinear methods—can effectively address the classification task.